

Video Vanguard

Also Remember...

- All video cards use a VGA output connector that allows the signal to be displayed on a monitor. Some cards supply and receive the video signal from television monitors, projectors and video cameras through optional coaxial or S-video interfaces.
- The feature connector exists on some cards and supports devices a TV tuner or a video capture board. A special connector is used to connect the video card to these additional devices through a flat ribbon cable.
- Memory bandwidth refers to the rate at which data is transferred between the graphics processor and graphics memory. If you have a card that utilises DDR SDRAM, operating at an effective rate of 301 MHz (say the GeForce DDR), you can calculate its memory bandwidth as follows:
 $128\text{-bit (16 bytes)} \times 301\text{ MHz} = 16 \times 301 = 4.81\text{ GBps}$
Compared to 460 MHz (for the GeForce2 ULTRA), where
 $128\text{-bit (16 bytes)} \times 460\text{ MHz} = 16 \times 460 = 7.36\text{ GBps}$
Compared to 125 MHz (for the TNT2 M64), where
 $64\text{-bit (8 bytes)} \times 125\text{ MHz} = 8 \times 125 = 1\text{ GBps}$
This clearly illustrates the difference that the memory speed causes on the total memory bandwidth available.

With the demise of 3dfx and with Matrox deciding to concentrate on 2D solutions, ATi and nVidia are the only high-end graphics card chip manufacturers remaining. Competition has surprisingly been not more in terms of technology but in terms of card manufacturers trying to outdo each other by trying to offer the best possible solution at a particular price point. Since ATi sells cards under its own brand name, the various card manufacturers primarily depend only on nVidia for their chipsets.

Mainstream desktop display solutions starting from Rs 1,500 to Rs 36,000 are now available in the market with a great number of manufacturers offering a wide variety of solutions in the low to mid segments. This was not the case in earlier times as you had to choose between cheap, low-powered solutions or very expensive models. However, with an increase in competition, the choices available have now improved significantly.

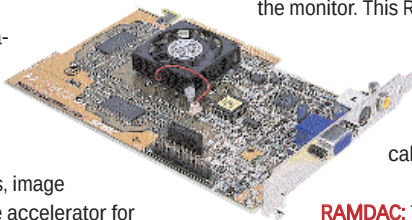
TECHNOLOGY INSIGHT

The main aim of a graphics accelerator is to process the 2D/3D elements associated with the GUI and applications such as a game or software. The execution of graphics involves the transfer of texture maps, image geometry, commands etc over to the accelerator for processing.

SMART TIPS

Graphics processor: The core of a graphics card, this is the component that actually processes the video information that is sent to it. Some of the best graphics processors in the market are the nVidia GeForce3, ATi Radeon 8500 and the 3dfxVoodoo 5 6000.

Video RAM: Video RAM on the graphics card is used to store the extensive graphical information like textures of the 3D objects. It is also used in the frame buffer, the part that holds the final frame that is to be displayed on



the monitor. This RAM is very similar to the one used as the main memory of your system, except that it is faster and optimised for working with video and graphical information.

RAMDAC: The Random Access Memory

Digital to Analog Converter is that chip on the graphics card which converts the digital information in the frame buffer (video memory) into a signal that can be displayed on your monitor. The RAMDAC on a graphics accelerator is specified by a frequency. The higher the frequency of the RAMDAC, the greater are the resolutions and the refresh rates that can be supported by the card.

Interface: The best choice available today is the AGP interface, which provides a very high-speed pathway for moving large quantities of graphic and image data between the graphics card and the rest of the system.

	Limited budget (support for decent refresh rates at low to medium resolutions)	Should run most games, but price is a consideration	Top-notch performance in 2D/3D, price no bar!
You need	A card that will run at 32-bit high colour at 1024x768 at 85 Hz with support for the odd game at low resolutions	A card that will give you a minimum of 30 fps in low to medium resolutions with a little compromise in quality	Maximum performance with all the eye candy enabled and no compromise on either resolution or quality
Look for	Cheaper or lower variants of faster chips. At least a card using the AGP interface with minimum 8 MB onboard RAM	A card with at least 16 MB or more memory running at 100 MHz or higher	A card with 32-64 MB RAM running at 166 MHz (preferably DDR); with deluxe features like VIVO and stereo glasses
Our pick	Galaxy Riva TNT2 M64	AOpen PA256 GeForce2 MX	ASUS AGP V7700 GeForce2 GTS, ASUS AGP V8200 GeForce3 Deluxe
Price range	Up to Rs 2,500	From Rs 2,500 to Rs 8,000	Rs 8,000 to Rs 36,000



The Best Products of the Year

December 2001
the di9it mark of excellence

